ERNESTINE HO

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EDUCATION

Toronto Metropolitan University (Formerly Ryerson University) ***Sep. 2020 – Jun. 2025***

Bachelor of Engineering – Computer Engineering

- Specialization in Software Engineering
- Dean's List Recipient – Fall 2024

SKILLS

Languages: Typescript/JavaScript, Python, Java, SQL, NoSQL, HTML/CSS, VBA, Kotlin

Frameworks: React, Next.js, Angular, Svelte, Node.js, FastAPI

Developer Tools: Git, GitHub Copilot, AWS IoT, Azure, Azure DevOps, GitHub Actions, PostgreSQL, MySQL, MongoDB, OpenAI APIs, Docker, Kubernetes, gRPC, Appium, LambdaTest

WORK EXPERIENCE

Software Developer Intern | Pomerleau ***Jan. 2024 – Sep. 2025***

- Revamped a React/TypeScript dashboard using AWS IoT Core and Kinesis Video Streams, supporting **5+ live feeds with sub-2-second latency**.
- Automated deployments via AWS CodePipeline and Docker on IoT Greengrass, **improving service uptime by 40%.**
- Built a Playwright test suite achieving **95% frontend coverage and cutting manual QA time by 50%**.

Software Developer Intern | LORA Technologies ***Jun. 2023 – Dec. 2023***

- Developed a full-stack AI financial chatbot with Svelte, FastAPI, and OpenAI APIs, **enabling natural language Q&A on financial data for 25+ internal users.**
- Automated verification of 100+ bot transactions using VBA and PostgreSQL queries, **cutting verification time by 36%.**
- Built Python validation workflows in Azure pipelines, **catching 10+ data errors per sprint** and ensuring PostgreSQL transactional accuracy.
- Wrote Java UI automation scripts with Appium and LambdaTest, **reducing weekly downtime by 4+ hours**.

PROJECTS

Personal Finance Tracker **Current**

- Building a full-stack personal finance tracker that ingests credit card PDF statements, classifies transactions using the **OpenAI API**, and stores them in **Supabase Postgres** via a **FastAPI** backend.
- Implemented **pgvector embedding-based merchant matching** to improve classification accuracy and reduce redundant API calls, with safeguards for rate limits and data validation.
- Developing a **Next.js** dashboard to visualize spending summaries and category breakdowns with real-time data aggregation.

Horizon Chase - Personal Finance Banking Application ***Jan. 2026***

- Built a full-stack banking application using **Next.js** and **TypeScript** with **MongoDB** for scalable data persistence and **Better Auth** for secure user authentication.
- Designed a responsive UI with **shadcn** component library and **Tailwind CSS**, following modern design principles.
- Deployed application on **Vercel** with **Sentry** monitoring and automated CI/CD pipeline for production reliability.

Social Networks-based Recommender System for Tourists (Capstone Project) ***Sep. 2024 – Apr. 2025***

- Developed a full-stack tourism recommender using **React/Typescript** and containerized **FastAPI** services with Docker, then deployed to **Azure Container Apps**.
- Built a **Databricks ETL pipeline** processing **3,800+ locations**, storing enriched results in **Azure Blob Storage**.
- Implemented a hybrid recommendation model with **OpenAI embeddings, scikit-learn** and **Surprise** models, improving **Top 10 accuracy by 18%**.